

UNIT IV – MULTIMODAL GENERATIVE AI APPLICATIONS

QUESTION 10 – AI-BASED RESEARCH ASSISTANCE

Engineering Research Topic Analysis and Summary

Professional Implementation Report

Question	Develop an AI-based Research Assistance application that accepts a research topic and generates relevant information, keywords, and a concise summary.
Folder Name	Q10_AI_Research_Assistant
Python File	research_assistant.py
Model	google/flan-t5-small (pre-trained language model)
Interface	Streamlit Web Application

1. QUESTION

Develop an AI-based Research Assistance application that accepts a research topic and generates relevant information, keywords, and a concise summary.

2. AIM

To develop an AI-based research assistant that accepts an engineering research topic and produces relevant technical information, useful research keywords, and a concise summary using a pre-trained language model.

3. OBJECTIVES

- Accept an engineering research topic from the user.
- Use a pre-trained language model to generate research-oriented content.
- Provide relevant information about the selected topic.
- Generate useful keywords for further literature search.
- Produce a concise summary that is easy for an engineering student to understand.
- Provide the output through an interactive Streamlit application.

4. SOFTWARE AND HARDWARE REQUIREMENTS

- Python 3.x
- Streamlit
- Transformers
- PyTorch
- SentencePiece
- Pre-trained google/flan-t5-small model
- Computer or laptop

5. FOLDER AND FILE STRUCTURE

```
UNIT_IV_Multimodal_Generative_AI
├── Q10_AI_Research_Assistant
│   └── research_assistant.py
```

6. TECHNOLOGIES USED

Technology	Purpose	Role
Python	Programming	Implements the research assistant
Streamlit	Web Interface	Provides topic input and result display
Transformers	NLP Framework	Loads and runs the pre-trained model
PyTorch	Model Runtime	Executes the language model
FLAN-T5 Small	Pre-trained Language Model	Generates the research information

7. SYSTEM WORKFLOW

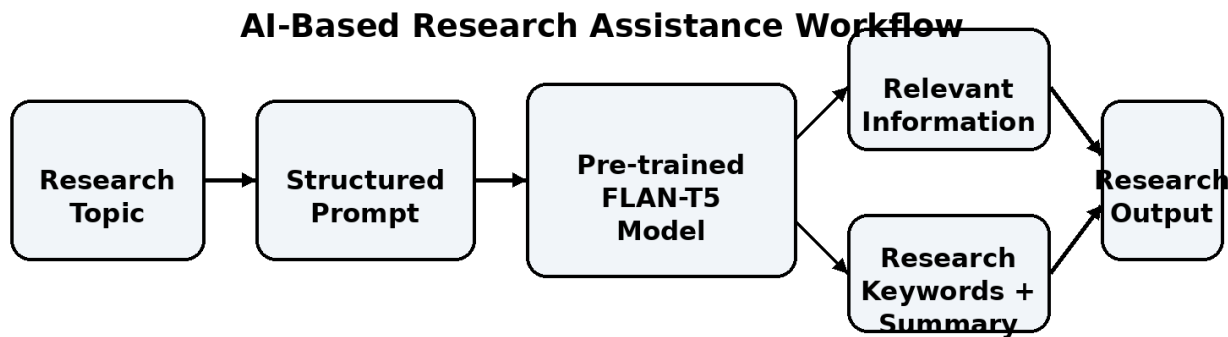


Figure 1. AI-based research assistance workflow.

The user enters a research topic. The application creates structured prompts for relevant information, keywords, and a concise summary. These prompts are processed by the pre-trained FLAN-T5 model, and the generated outputs are displayed in the Streamlit interface.

8. STEP-BY-STEP IMPLEMENTATION

- Step 1: Create the Q10_AI_Research_Assistant folder.
- Step 2: Create the research_assistant.py file.
- Step 3: Create and activate the Python environment.
- Step 4: Install Streamlit, Transformers, PyTorch and SentencePiece.
- Step 5: Load the pre-trained google/flan-t5-small tokenizer and model.
- Step 6: Create a text box for entering the engineering research topic.
- Step 7: Validate the topic before processing.
- Step 8: Construct separate prompts for relevant information, keywords and concise summary.
- Step 9: Tokenize each prompt.
- Step 10: Generate the response using the pre-trained model.
- Step 11: Decode the generated tokens into readable text.
- Step 12: Display the three research outputs in the Streamlit application.
- Step 13: Provide a download option for the combined research output.

9. COMPLETE IMPLEMENTATION CODE

```

import streamlit as st
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

st.set_page_config(
    page_title="AI Research Assistant",
    page_icon="📖",
    layout="wide"
)

```

```

st.title("🔗 AI-Based Engineering Research Assistant")

st.write(
    "Enter an engineering research topic to generate "
    "relevant information, keywords, and a concise summary "
    "using a pre-trained language model."
)

@st.cache_resource
def load_model():
    model_name = "google/flan-t5-small"

    tokenizer = AutoTokenizer.from_pretrained(
        model_name
    )

    model = AutoModelForSeq2SeqLM.from_pretrained(
        model_name
    )

    return tokenizer, model

tokenizer, model = load_model()

def generate_response(prompt):

    inputs = tokenizer(
        prompt,
        return_tensors="pt",
        truncation=True,
        max_length=512
    )

    outputs = model.generate(
        **inputs,
        max_new_tokens=220,
        num_beams=4,
        early_stopping=True
    )

    return tokenizer.decode(
        outputs[0],
        skip_special_tokens=True
    )

topic = st.text_input(
    "Research Topic:",
    value="Artificial Intelligence in Smart Manufacturing"
)

```

```

if st.button("🔍 Generate Research Information"):

    if not topic.strip():
        st.warning("Please enter a research topic.")
    else:

        information_prompt = f"""
You are an engineering research assistant.

Explain:
{topic}

Include meaning, major concepts,
engineering applications, importance,
and practical relevance.
Keep the response concise.
"""

        keyword_prompt = f"""
Generate important research keywords
for the engineering topic:
{topic}

Return concise keywords.
"""

        summary_prompt = f"""
Write a short meaningful summary
for an engineering student about:
{topic}
"""

        information = generate_response(
            information_prompt
        )

        keywords = generate_response(
            keyword_prompt
        )

        summary = generate_response(
            summary_prompt
        )

        st.success(
            "Research information generated successfully!"
        )

        st.subheader("📄 Relevant Information")
        st.info(information)

```

```
st.subheader("🔗 Research Keywords")
st.success(keywords)
```

```
st.subheader("📄 Concise Summary")
st.warning(summary)
```

```
complete_output = f"""
RESEARCH TOPIC
{topic}
```

```
RELEVANT INFORMATION
{information}
```

```
RESEARCH KEYWORDS
{keywords}
```

```
CONCISE SUMMARY
{summary}
"""
```

```
st.download_button(
    label="📄 Download Research Output",
    data=complete_output,
    file_name="research_assistance_output.txt",
    mime="text/plain"
)
```

10. EXECUTION COMMAND

```
streamlit run ".\Q10_AI_Research_Assistant\research_assistant.py"
```

11. TEST RESEARCH TOPIC

Artificial Intelligence in Smart Manufacturing

12. ACTUAL IMPLEMENTATION RESULTS

The screenshots below show the actual working local implementation. The pre-trained language model loaded successfully, the research topic was accepted, and the application generated separate sections for relevant information, research keywords, and a concise summary.

AI-Based Engineering Research Assistant

Enter an engineering research topic to generate relevant information, keywords, and a concise summary using a pre-trained language model.

Pre-trained language model loaded successfully.

Enter Research Topic

Research Topic:

Artificial Intelligence in Smart Manufacturing

 Generate Research Information

Research information generated successfully!

Figure 2. Q10 application showing successful model loading, research-topic input and successful generation.

Relevant Information

Artificial Intelligence in Smart Manufacturing is a tool used in manufacturing automation.

Research Keywords

Artificial intelligence in smart manufacturing

Concise Summary

Artificial Intelligence in Smart Manufacturing

 Download Research Output

Figure 3. Q10 application showing Relevant Information, Research Keywords and Concise Summary.

13. OBSERVED OUTPUT

Output Section	Observed Result	Purpose
Relevant Information	Artificial Intelligence in Smart Manufacturing is a tool used in manufacturing automation.	Provides a brief technical explanation.
Research Keywords	Artificial intelligence in smart manufacturing	Provides a topic-related search term.
Concise Summary	Artificial Intelligence in Smart Manufacturing	Provides a short summary output.

14. RESULT

The AI-Based Engineering Research Assistant was successfully implemented and tested using the research topic “Artificial Intelligence in Smart Manufacturing”. The local pre-trained FLAN-T5 model loaded successfully and the application generated all three required output categories: relevant information, research keywords, and a concise summary.

15. ADVANTAGES

- Uses a pre-trained language model without requiring API credits.
- Provides three research-oriented outputs from one topic.

- Simple and interactive Streamlit interface.
- Useful for initial project, seminar and literature-study preparation.
- Can run locally after the model has been downloaded and cached.

16. LIMITATIONS

- The small FLAN-T5 model may generate very short or incomplete research content.
- Generated content should be verified against reliable academic sources.
- The model does not automatically retrieve research papers or citations.
- Output quality depends on the prompt and model capabilities.

17. FUTURE ENHANCEMENTS

- Integrate research-paper retrieval and citation generation.
- Use RAG to ground answers in academic documents.
- Add research-gap and objective generation.
- Add literature-review support.
- Add multilingual research assistance.

18. CONCLUSION

The developed AI-Based Research Assistant demonstrates how a pre-trained language model can support engineering research activities. The system accepts a research topic and produces relevant information, keywords and a concise summary through a simple Streamlit interface. The successful local execution confirms that the required application has been implemented without relying on paid inference credits.

19. VIVA / KEY POINTS

- The application accepts an engineering research topic.
- FLAN-T5 is used as the pre-trained language model.
- The system generates relevant information, keywords and a concise summary.
- Streamlit provides the interactive user interface.
- The application can be enhanced with RAG and academic-paper retrieval.